

Bachelor of Technology — General Engineering - AI and Robotics

Table 1: General Engineering - AI and Robotics — B23 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	39
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	37
	Discipline Electives	29
Electives	Free Electives	22
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

Table 2: B. Tech General Engineering - AI and Robotics — B23 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	DS-201	Data Handling and Vizualization	2-0-2-3	3
2	EE-261	Electrical System Around Us	3-0-0-3	3
3	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
4	EE-201	Electromechanics	2.5-0.5-0-3	3
5	IC-253	Programming and Data Structures	2-0-2-3	3
6	ME-100	Reverse Engineering	0-0-2-1	1
7	AR-520	Design Practicum of Mechatronic Systems	2-0-2-3	3
8	AR-521	Control of Robotic Systems		3
9	AR-523	Robot Manipulators		3
10	EE-302	Control Systems	3-0-2-4	4
11	ME-305	Design of Machine Elements	3-1-0-4	4
12	ME-309	Theory of Machines	3-1-0-4	4
Total				37

Table 3: B. Tech General Engineering - AI and Robotics — B23 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket	IC-230	Environmental Science	3-0-0-3	3
6	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
7		HS-342	German I	3-0-0-3	
8		HS-108	Basic English for Engineers	3-0-0-3	
Total					

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 4: B. Tech General Engineering - AI and Robotics — B23 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-240	Mechanics of Rigid Bodies	1.5-1.5-0-3	3
9	IKS + HSS (choose 0-3 credits)	IC-181	Indian Knowledge Systems	2-0-2-3	0-3
10		HS-151	Introduction to English Literature	3-0-0-3	
11		HS-357	French Language and Culture	2-0-0-2	
12		HS-308	European Literature	2-0-0-2	
Total					22-25

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 5: B. Tech General Engineering - AI and Robotics — B23 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	DS-201	Data Handling and Vizualization	2-0-2-3	3
4	DC	EE-261	Electrical System Around Us	3-0-0-3	3
5	DC	ME-206	Mechanics of Solids	2.5-0.5-0-3	3
Total					15

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 6: B. Tech General Engineering - AI and Robotics — B23 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	EE-201	Electromechanics	2.5-0.5-0-3	3
2	DC	IC-253	Programming and Data Structures	2-0-2-3	3
3	DC	ME-100	Reverse Engineering	0-0-2-1	1
Total					7

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 7: B. Tech General Engineering - AI and Robotics — B23 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	AR-520	Design Practicum of Mechatronic Systems	2-0-2-3	3
2	DC	AR-521	Control of Robotic Systems		3
3	DC	AR-523	Robot Manipulators		3
4	DC	EE-302	Control Systems	3-0-2-4	4
5	DC	ME-305	Design of Machine Elements	3-1-0-4	4
6	DC	ME-309	Theory of Machines	3-1-0-4	4
Total					21

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 8: B. Tech General Engineering - AI and Robotics — B23 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	ISTP	DP-301P	Interdisciplinary Socio-Technical Practicum	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 9: B. Tech General Engineering - AI and Robotics — B23 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	GE-498P	Major Technical Project - I	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 10: B. Tech General Engineering - AI and Robotics — B23 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-010	Internship		2
2	MTP	GE-499P	Major Technical Project - II	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 29 credits of **Discipline Electives** and 22 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 11: General Engineering - AI and Robotics — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	AR-502	Advanced Design Practicum	4
2	AR-505	Principles of Robot Autonomy	3
3	AR-508	Marine Robotics	3
4	AR-509	Deep Learning for Robotics	4
5	AR-510	Underactuated Robotics	3
6	AR-511	Autonomous Mobile Robots	3
7	AR-513	Unmanned Aerial Systems (UAS)	3
8	AR-514	Vision and Learning Based Control	3
9	AR-519	Robot Manipulators	3
10	AR-522	Robot Vision	3
11	AR-524	Robot Simulators	3

General Engineering - AI and Robotics — Available Discipline Electives (choose from these)
(continued)

Sl. No	Course Code	Course Name	Credits
12	AR-525	Reinforcement Learning for Robotics	3
13	AR-526	Multimodal Learning	3
14	CE-251	Hydraulics Engineering	3
15	CS-212	Design and Analysis of Algorithms	4
16	CS-514	Advance Data Structure and Algorithm (CS-580)	4
17	DS-314	Probability Theory and Statistics	4
18	EE-201P	Electromechanics Lab	1
19	EE-203	Network Theory	3
20	EE-211	Analog Circuit Design	3
21	EE-260	Signals and Systems	3
22	EE-529	Embedded Systems	4
23	EE-604	Practicum on Advanced Electric Drives	3
24	EE-608	Digital Image Processing	4
25	IC-241	Materials Science for Engineers	3
26	ME-210	Fluid Mechanics	3
27	ME-212	Product Manufacturing Technology	3
28	ME-213	Engineering Thermodynamics	4
29	ME-510	Advanced Manufacturing Processes	3
<i>Compulsory project components — counted as DE, not selectable</i>			
30	GE-498P	Major Technical Project - I (compulsory; counted as DE)	4
31	GE-499P	Major Technical Project - II (compulsory; counted as DE)	4